ADG classes: Randomized motif finding

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Today we will consider the *planted motif search* problem. Given a set of input sequences, the planted (l, d)-motif search problem is to find all strings x of length l such that every input sequence contains a substring with Hamming distance to x of at most d.

1 Planting motifs in sequences

1. Inspect the code for generating planted motif problem instances.
2. Complete the code for `planted_motif_instance()` so that it returns a list of t random sequences of length n, each with planted one occurrence of the mutated motif. The mutation should be performed on d positions.

2 Gibbs sampler

1. Complete the code for `consensus()` to compute the consensus sequence of a list of l-mers.
2. Write the function `weighted_choice()` that uses arbitrary weights. Hint: standard library function `random.choices()` can take argument `weights`.
3. Complete the code for `gibbs_sampler()` to implement the sampling as in the lecture slides.

3 Random projection using gapped k-mers

1. Adapt the code of `simple_random_projection()` to implement the full `random_projection()`. In particular, use the Gibbs sampler to refine the initial motif, and repeat the procedure for multiple initial motifs.

2. Discuss if the l-mers need to be stored in the buckets in the function `simple_random_projection()`.

3. Test the algorithms on the planted $(15, 4)$-motif search problem, trying different parameter settings.